9–THE INCLINING EXPERIMENT

9.1 PURPOSE OF EXPERIMENT

In theory it should be possible to determine the position of the lightship vertical centre of gravity by taking moments about the keel of all the steel, welding electrodes, machinery, equipment and fixtures that go into building a vessel. This will however be tedious and the results will be of doubtful accuracy.

The alternative is therefore to conduct an Inclining Experiment on a ship when construction is completed to determine the position instead.

9.2 CONDUCT OF EXPERIMENT

The experiment is conducted by moving a known mass or masses a known distance transversely across a vessel to deliberately cause a list and the shift in centre of gravity (C of G) is then calculated. From the results so obtained the lightship metacentric height is determined and thence the vertical C of G.

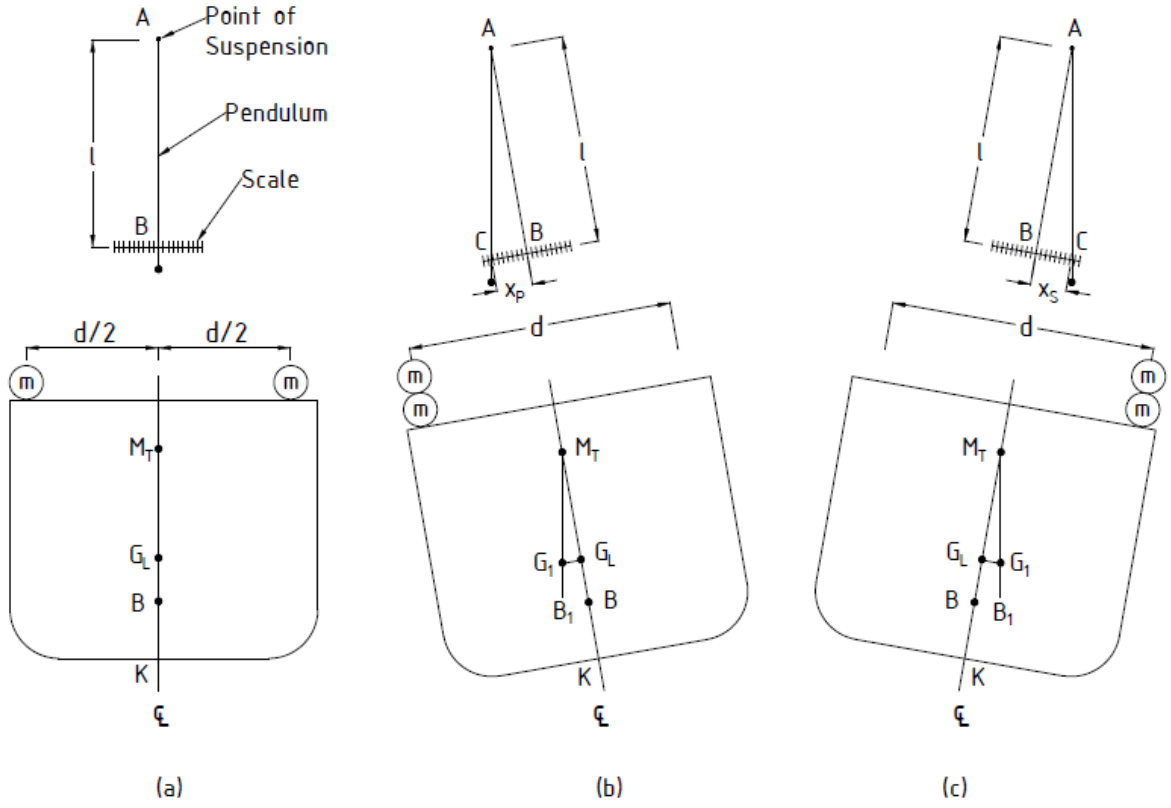


Figure 9.1 – Conduct of Experiment

Procedure:

1. Lay a track across the vessel on which the mass (m) or masses (m) will be moved.
2. Rig a pendulum with length (l) and a scale arrangement as in Figure 9.1(a).
3. Starting with the vessel upright place the two masses at equal distances from the centreline as in Figure 9.1(a).
4. Move the starboard side mass to port to cause a list as in Figure 9.1(b).

5. Read and record the deflection (x_P) from the scale.
6. Return the mass to its original position to bring the vessel back upright.
7. Repeat the experiment by shifting the port side mass to starboard as in Figure 9.1(c) and record the deflection (x_S).
8. Take the following readings to complete the experiment:
    a. Fore and aft draughts when vessel is upright.
    b. Density of water computed from samples collected over a wide area.

Calculations:

In triangles $G_LG_1M_T$ and ABC, by the property of similar triangles:

$$\tan\theta = \frac{G_LG_1}{G_LM_T} = \frac{x}{l}$$

From which:

$$GM_T = \frac{G_LG_1 \times l}{x}$$

Where:

$$x = \frac{x_P + x_S}{2}$$

Now:

$$G_LG_1 = \frac{m \times d}{\Delta_L}$$

Therefore

$$GM_T = \frac{mdl}{\Delta_L x}$$

Determine the mean draught from the fore and aft draughts recorded earlier and use the result to obtain the values of ∇_L and KM_T from the Hydrostatic Curves of the ship.

Determine the average density and multiply it by ∇_L to obtain Δ_L used in the calculation of GM_T.

Finally calculate:

$$KG_L = KM_T - GM_T$$

9.3 PRECAUTIONS DURING EXPERIMENT

Ideally the experiment must be completed when construction is completed and the ship is absolutely empty. However during the experiment there are usually pieces of items of relatively small masses, such as staging, still remaining on board, or tools, stores, consumables, etc. that are yet to be loaded. Their masses and centres must therefore be ascertained and the necessary corrections made thereafter.

To ensure accurate results the following additional precautions are necessary during the conduct of the experiment:

1. The weather must be as calm as possible.
2. Mooring lines must be as slack as may be allowed.
3. All loose weights must be lashed or secured.
4. Only persons involved in the experiment are to be allowed on board.
5. Persons allowed on board are to stay on the centerline throughout the experiment.
6. Use at least two pendulums placed wide apart and of different lengths to determine the deflection. This way the observers will be independent of each other, and the problem of human error or fixing will be avoided or minimized.
7. Series of readings are to be taken and thereafter averaged to compensate for continuously changing weather conditions which are beyond human control.